

Vivado HLS 2019.1 - Syntax

Optimisation

Optimisation flags are used with the key word `#pragma HLS INTERFACE` to call Vivado's optimisation behaviors

```
#pragma HLS INTERFACE ap_ctrl_none port=return           // disable
unwanted messages
#pragma HLS INTERFACE axis register both port=output_stream //
#pragma HLS INTERFACE axis register both port=input_stream //
```

Details are given in the [Vivado HLS 2019.1 - HLS Pragma Optimizations](#).

Loop Naming

It is possible to give a name to each loop line with a `flag_name: ...` before the code.

```
// naive matrix multiplication algorithm
loop_mult_row: for(row = 0; row < SIZE; row++){
    loop_mult_col: for(col = 0; col < SIZE; col++){
        float sum = 0; // same type as matrix
        loop_mult_k: for(k = 0; k < SIZE; k++){
            sum += A[row][k] * B[k][col];
        }
        C[row][col] = sum;
    }
}
```